Effects of Semantic Distance on Suppressed Concept Representations

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Seal or Signature

1. Summary of the Research

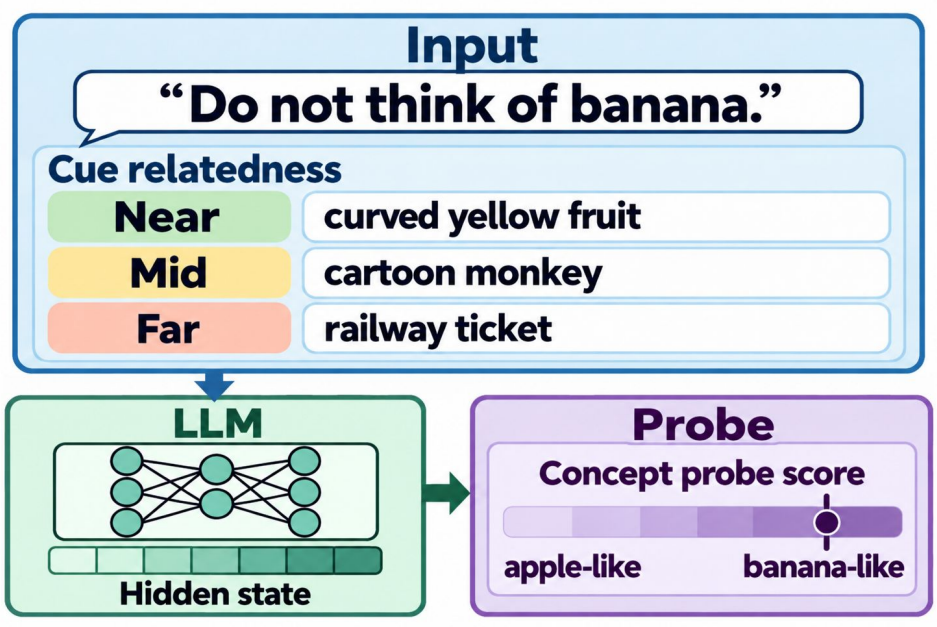
Large language models (LLMs) can be instructed not to think about a particular concept. However, avoiding a word in the final response does not necessarily mean that information about the concept disappears from the model's internal representation. Mann et al. [1] investigated thought suppression in LLMs under distracting contexts, while Zhou et al. [2] examined the internal mechanisms of negation. Building on these studies, we investigate how related information affects the internal representation of a suppressed concept, focusing on the relationship between semantic distance and changes in internal representations.

2. Approach / Methodology

Large language models (LLMs) can be instructed not to think about a particular concept. However, avoiding a word in the final response does not necessarily mean that information about the concept disappears from the model's internal representations. Mann et al. [1] investigated thought suppression in LLMs under distracting contexts, while Zhou et al. [2] examined how language models internally process negation.

Building on these studies, we investigate how related information affects the internal representation of a suppressed concept. Specifically, we focus on the relationship between semantic distance and changes in internal representations. Our goal is to determine whether concept-related information remains detectable even when the model is instructed to suppress it.

Fig. 1 Proposed Experimental Framework



3. Current Results and Status

Preliminary experiments were conducted using eight target concepts. We examined target-word mentions under different instruction and cue conditions. We also trained concept probes to examine internal representations. In a small validation set, the final-layer probe achieved a mean ROC-AUC of 0.972, suggesting that information distinguishing the concepts could be extracted from hidden states.

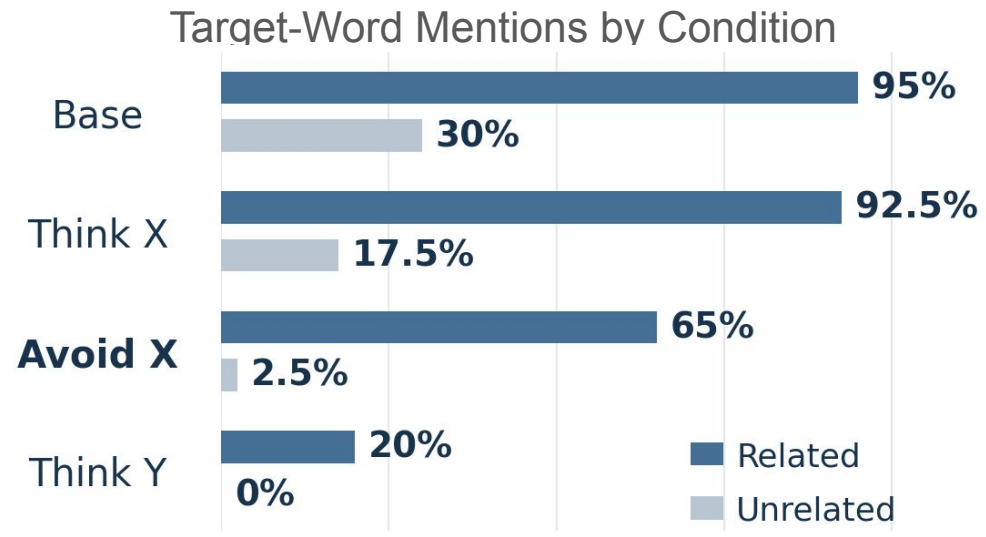


Fig. 2 shows that related cues increased target-word mentions compared with unrelated cues across the tested instruction conditions. Under suppression, mention rates were 65% and 2.5%, respectively. However, suppression still reduced mentions compared with neutral instructions (95% with related cues). These preliminary results motivate further investigation of semantic distance and internal representations.

4. Remaining Tasks and Schedule

- **Oct** Prepare semantic-distance cues
- **Nov** Analyze hidden state effects
- **Dec** Compare experimental conditions
- **Jan** Summarize results & write thesis

References

[1] L. Mann et al., “Don't Think of the White Bear: Ironic Negation in Transformer Models Under Cognitive Load,” *arXiv:2511.12381*, 2025.

[2] Z. Zhou et al., “How Language Models Process Negation,” *Proc. ICML*, PMLR 306, pp. 166474–166494, 2026.